

Phase Diagram

❖ Role of the phase diagram :-

- Shows the phases in equilibrium for a given alloy composition, at a glance.
- Shows the relationship between the composition , temperature, and alloy structure in series.
- There is a strong correlation between microstructure and mechanical properties, and the development of microstructure of an alloy is related to the characteristics of its phase diagram.
- Phase diagrams provide valuable information about melting, casting, crystallization, phase separation ,solidification of metals and alloys , purification of materials and other phenomena.

Classification Of Phase Diagram

- **Unary (or one component) phase diagram** which is plotted as pressure on the vertical axis and temperature on the horizontal axis.
- **Binary (or two components) phase diagram** which is plotted with temperature taken on the vertical axis and the various concentrations taken on the horizontal axis commonly.
- **Ternary (or three components) phase diagram** which is plotted three dimensionally between three components and the equilibrium associated with each other and the vertical axis usually taken comprises of the temperature as the common parameter.

Phase Diagram- Basic Terms

❖ System

- The substances that isolated and unaffected by their surrounding are known as system.
- It may be composition of solid, liquid , gases or the combinations and may have metals and nonmetals separately or in any combination.
- A system is capable of changing its composition, temperature, pressure, density etc.

❖ Variables

- A particular phase exists under various condition of pressure and temperature and composition. These parameters are known as the variables of the phase.

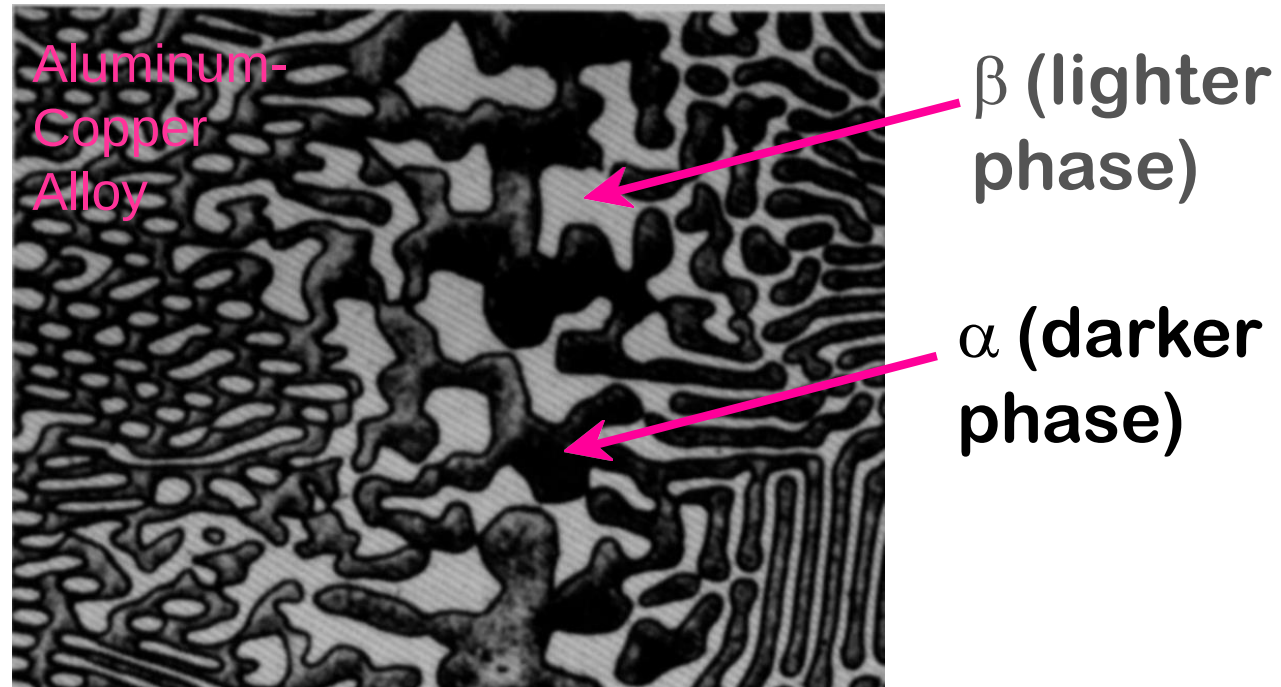
❖ Components

- These are the substances, element or chemical compound whose presence is necessary and sufficient to make a system.
- A pure metal is one component system whereas and alloy of metals is a two-component(binary) system etc.
- Components are the elements or compounds that are mixed initially (Example: Al and Cu).

Phase Diagram- Basic Terms

❖ Phases

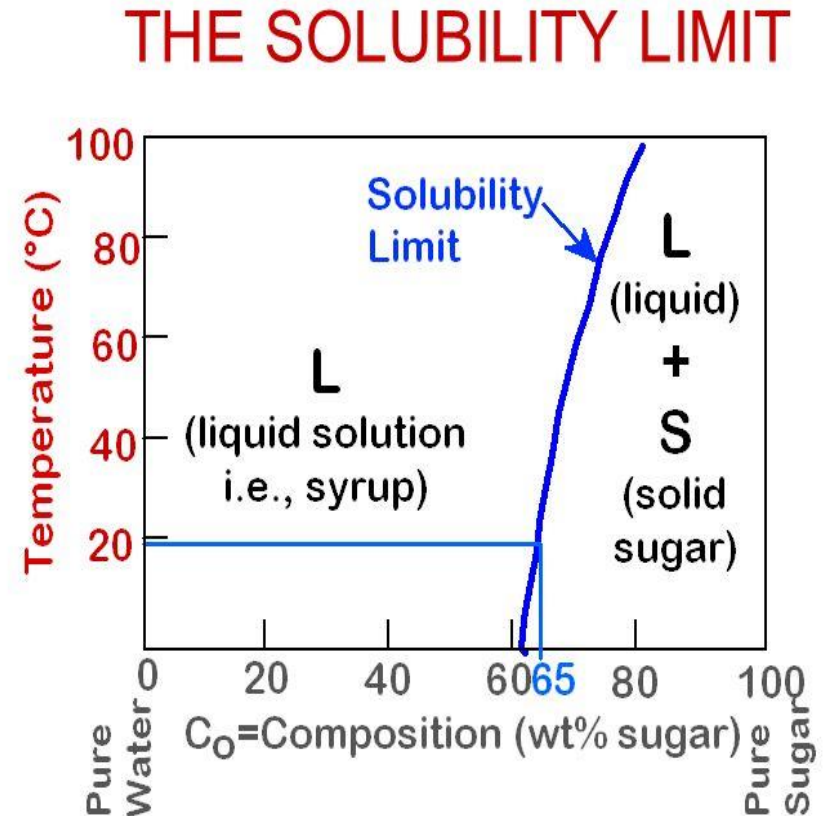
- A phase may be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics, and mechanically separable portion of the material with a given chemical composition and structure .
- Every pure material is considered to be a phase.
- In a equilibrium diagram, liquid is one phase and solid solution is another phase.



Phase Diagram- Basic Terms

❖ Solubility limit

- For many alloy systems and at some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution; this is called a solubility limit.
- The addition of solute in excess of this solubility limit results in the formation of another solid solution or compound that has a distinctly different composition.

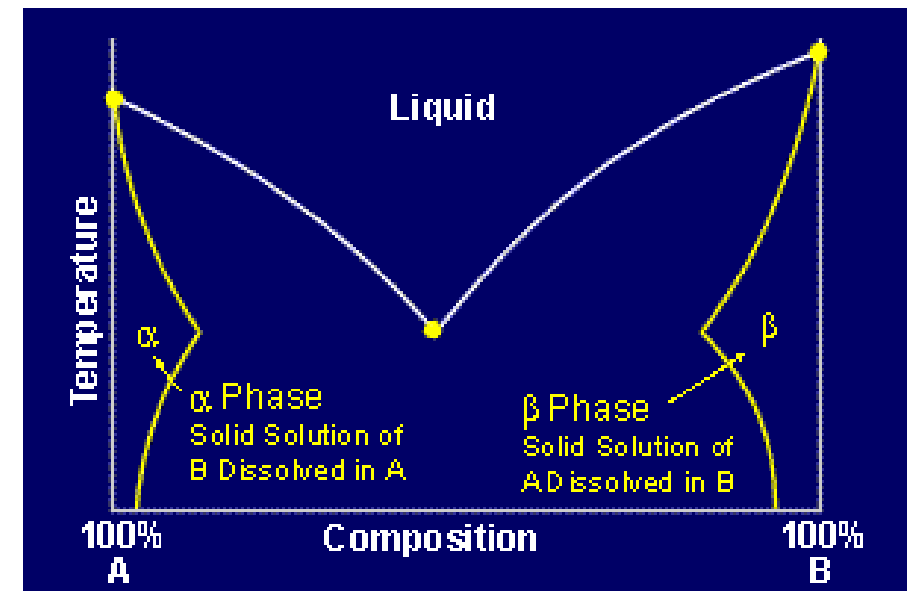
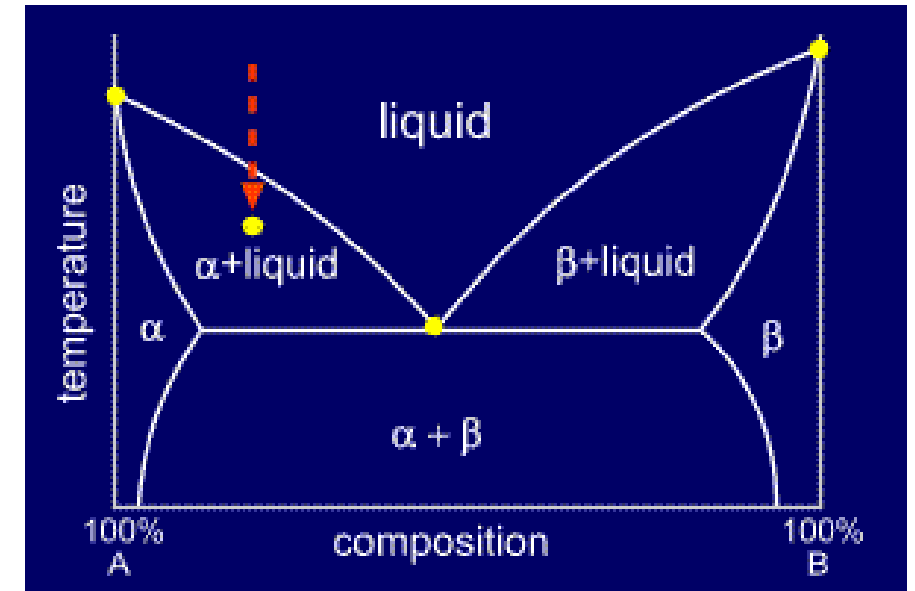


- Ex: Phase Diagram: Water-Sugar System

Phase Diagram- Basic Terms

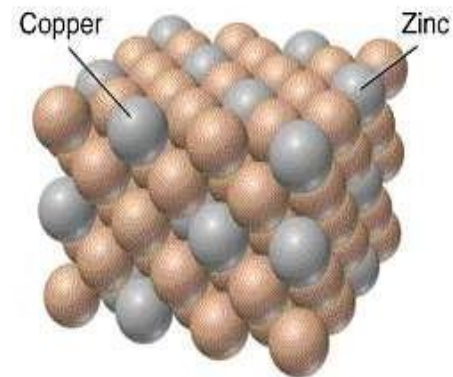
❖ Alpha and beta phases

- The alpha and beta phases are phases that represent a solid solution of elements A and B.
- The alpha phase will have a crystal structure like pure A and will contain some B substituting in the crystal lattice.
- Likewise, beta phase will have an element B crystal structure with A substituting for some B atoms.

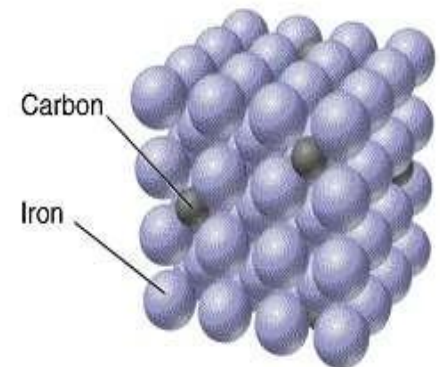


Solid Solutions

- A solid solution is defined as a solid mixture containing a minor component uniformly distributed within the crystal lattice of the major component. (as shown in the two possible cases)
- The key feature of a solid solution is that the metals retain their homogeneity and hence their solubility after their transformation to a solid crystalline state.
- An example of a solid solution would be Copper 64% and Zinc 36%
- The solid solutions comprise of atoms of almost the same atomic radius , and they tend to form a single phase and the elements are soluble in both their solid and liquid states .



A Brass, a substitutional alloy



B Carbon steel, an interstitial alloy

Solid Solutions

- **The element present in the alloy in the largest portion is referred as base metal or parent metal or solvent and the other elements are referred as alloying element or solute.**
- **Solid solution is a type of alloy in which the atoms of alloying element are distributed in the base metal and both have similar crystal structure.**
- **The composition of alloying element may vary but the structure should be similar to base metal.**

Classification Of Solid Solutions

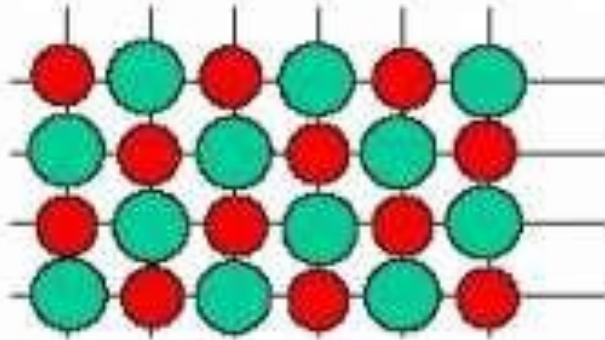
❖ **Substitutional – Regular or Ordered & Random or Disordered**

❖ **Interstitial**

Ordered Substitutional and Interstitial Compounds

Substitutional

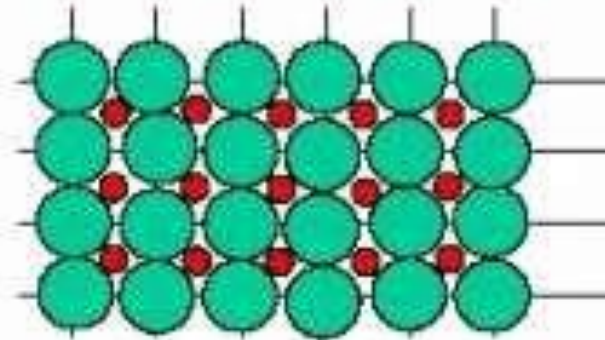
'an element replaces host atoms in an orderly arrangement'



e.g., Ni_3Al (hi-T yield strength),
 $\text{Al}_3(\text{Li}, \text{Zr})$ (strengthening)

Interstitial

'an element goes into holes in an orderly arrangement'



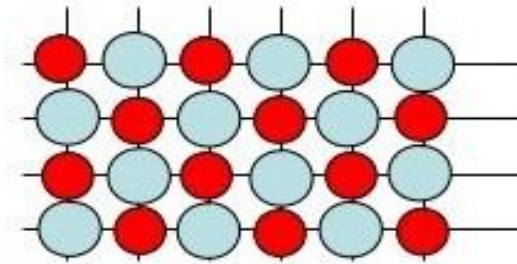
e.g., small impurities, clays
ionic crystals, ceramics.

Substitutional solid solutions

- Substitutional solid solutions have a direct substitution of one type of atom for another so that solute atoms (Cu) enter the crystal to take positions normally occupied by solvent atoms (Eg. Ni atoms).
- Ordered substitutional solid solutions are those in the disordered solution is cooled slowly, causing atomic re-arrangement because of the diffusion due to cooling causing uniform distribution of solute and solvent atoms.

Ordered Substitutional solid solution

Substitutional
element replaces host atoms
in an orderly arrangement



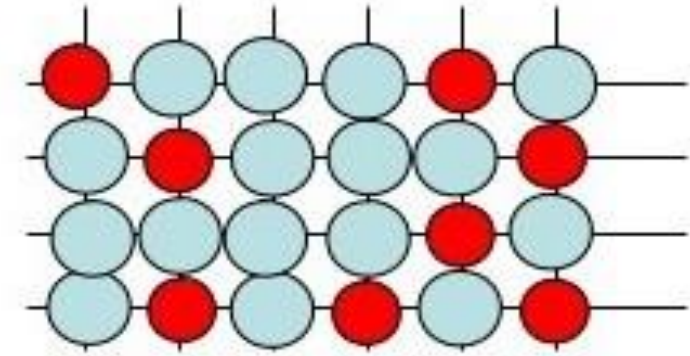
e.g., Ni-Al, Al-(Li,Zr)

● Ni (solvent)
● Al (solute)

Substitutional solid solutions

Disordered substitutional solid solutions are those type of solid solutions in which the solute atoms are randomly distributed in the solvent lattice structure , hence disordered.

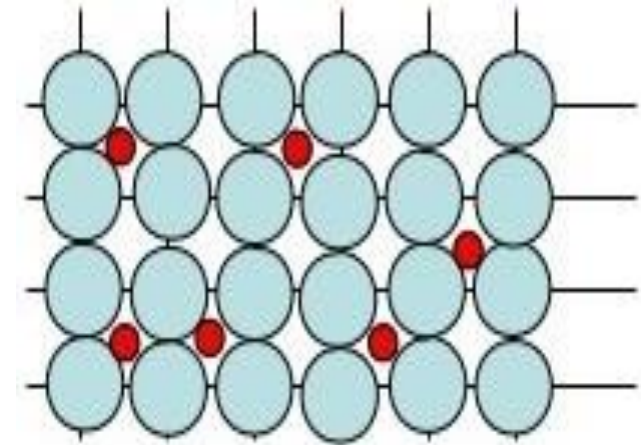
Example: Alpha brass



● Copper solvent
● Zinc solute

Interstitial Solid Solution

Interstitial solid solutions are those in which the solute atoms occupy the interstitial positions in the solvent crystal lattice, without any replacement of solvent atoms, on account of a very small solute atomic size relative to that of the solvent. Example: Fe-C



● Iron (solvent)

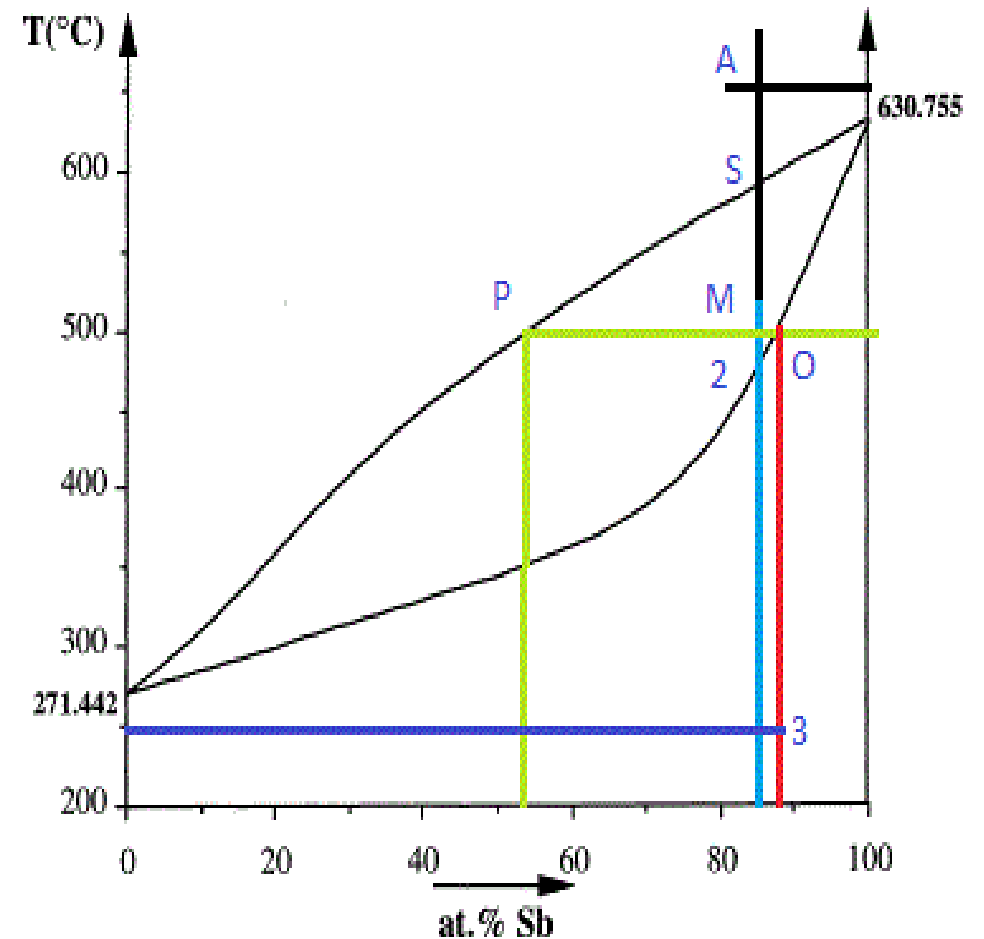
● Carbon (solute)

Equilibrium

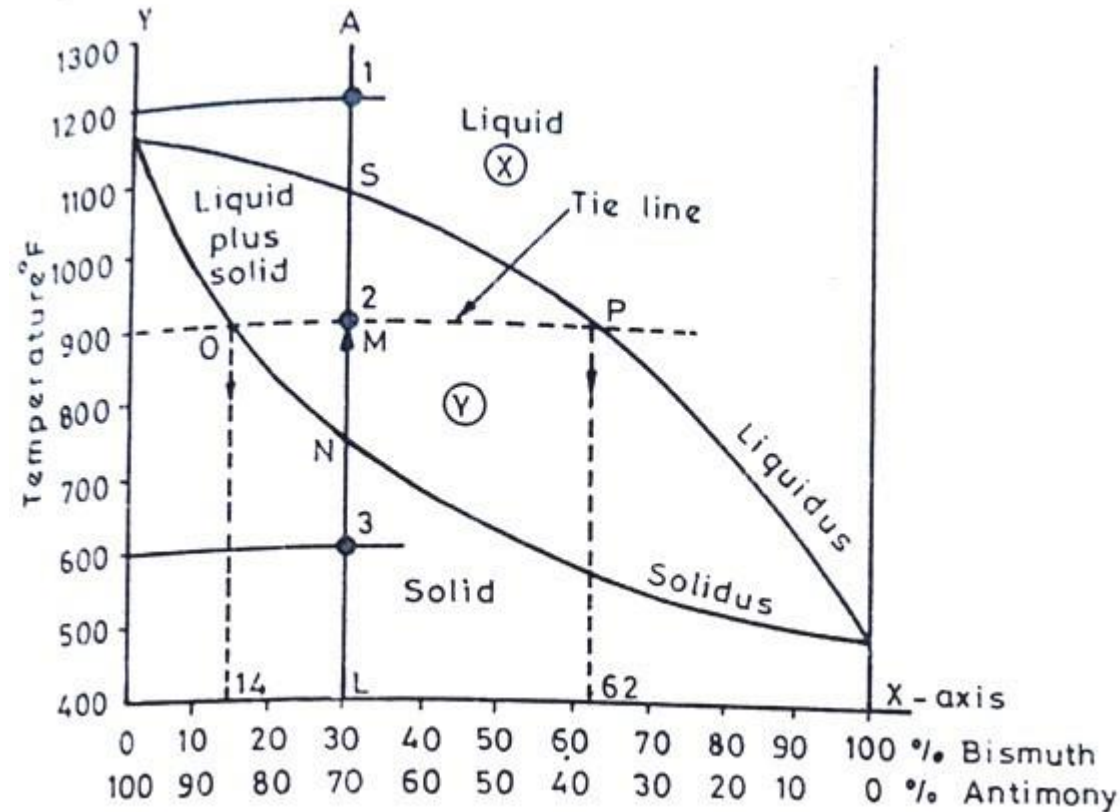
- **A system is at equilibrium if its free energy is at a minimum at a specified combination of temperature, pressure and composition.**
- **The (macroscopic) characteristics of the system do not change with time — the system is stable.**
- **A change in temperature(T), pressure (P) or composition (C) for the system will result in an increase in the free energy and possible changes to another state whereby the free energy is lowered.**

Phase Rules

- The following are the rules which are to be followed while interpreting the phase diagrams :
- ❖ Prediction of Phases
 - From a phase diagram , specific information can be obtained only if a temperature and a composition is specified.
 - This can be done by using both temperature and composition parameters to identify the point on the phase diagram.
 - Once done , depending on its location ,identify the phase present at that particular point.
- ❖ Phase composition
 - To find the phase composition at a particular temperature like 500 degrees Celsius , draw a horizontal line OP , that hits the liquidus and a vertical line that hits the solidus curve from P .
 - This would automatically indicate the phase composition for the required temperature.



Interpretation of phase diagram

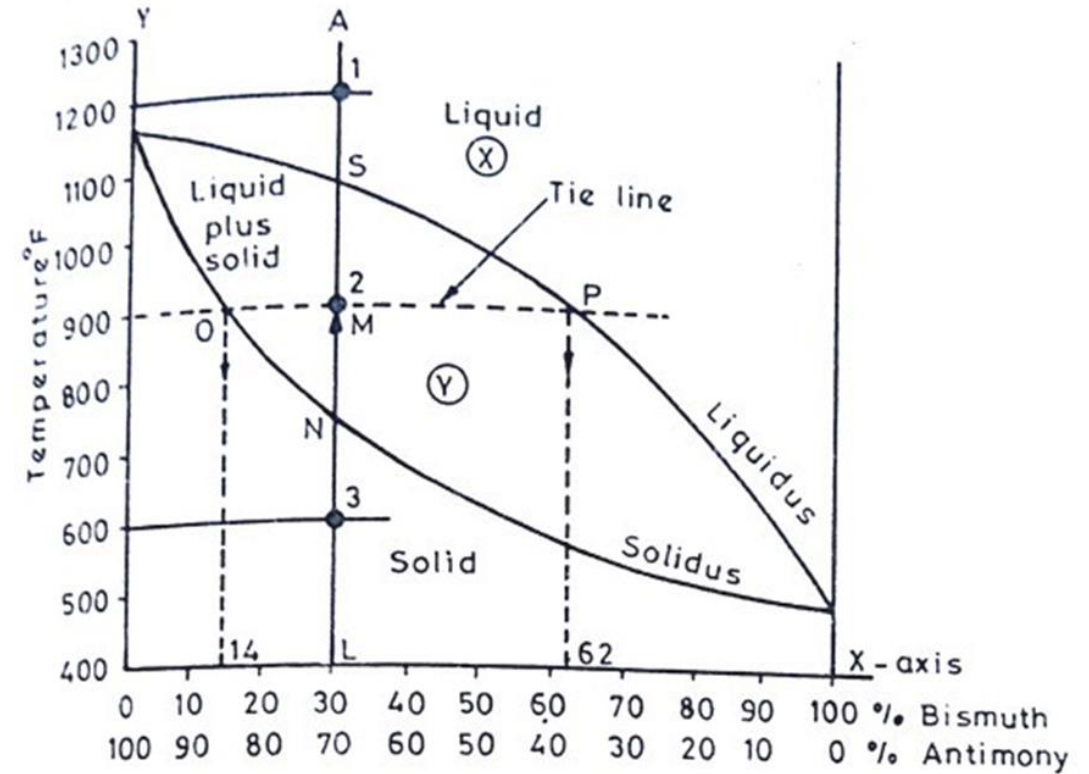


Following the three useful conclusions are the rules necessary for interpreting phase diagram.

- Rule -1 : Prediction of phases
- Rule -2 : Phase Composition
- Rule -3 : Lever arm principle

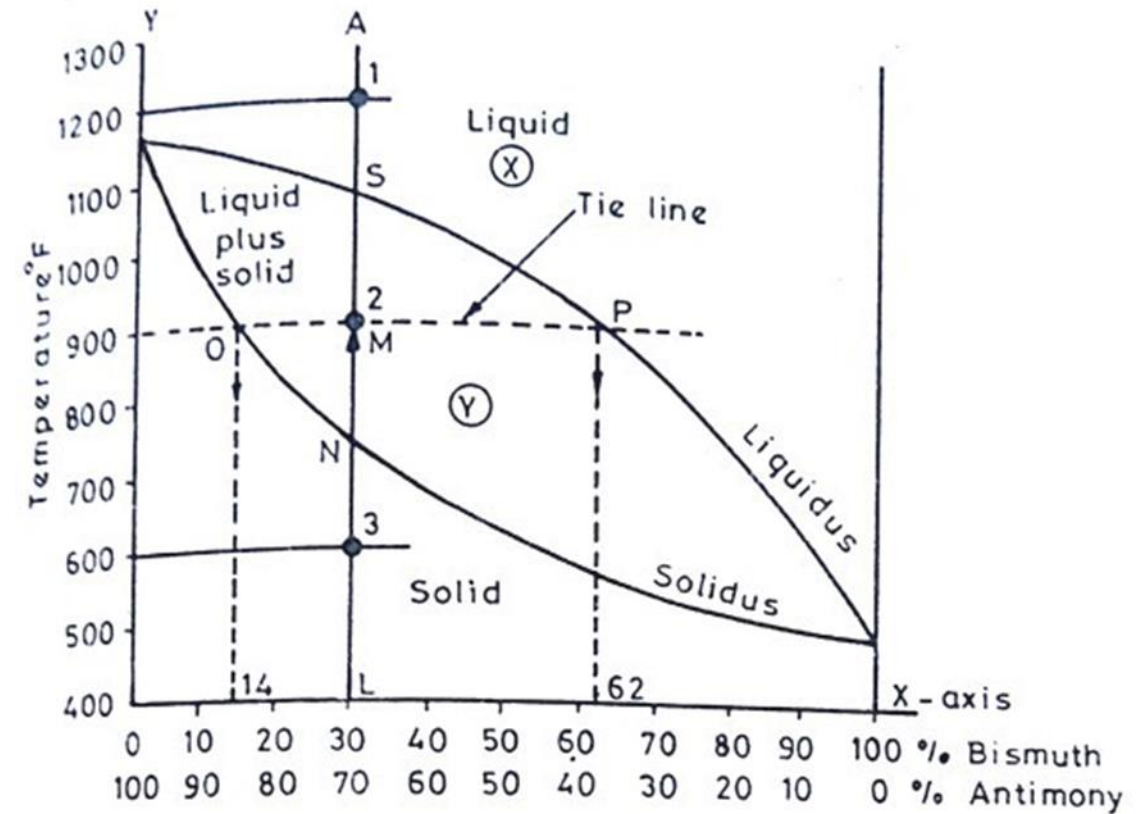
Rule -1 : Prediction Of Phase

- Form a phase diagram, specific information can be obtained only if a temperature and a composition are specified.
- For example, the state of the alloy of composition 30% bismuth can be determined only with reference to a certain temp.
- Thus when this alloy is at 1200°F, point 1 located and when it is at 900°F and 600°F, points 2 and 3 are located respectively.
- The next step is to determine the phase or phases present at points number 1, 2 and 3.
- Point -1 : with 30% Bi-70%Sb alloy at 1200°F, only one phase, i.e., the liquid solution is present.
- Point -2 : with the same alloy, but 900°F, two phases are present, i.e., liquid solution and solid solution.
- Point -3 : with the same alloy, but 600°F, only one phase, i.e., the solid solution is present.

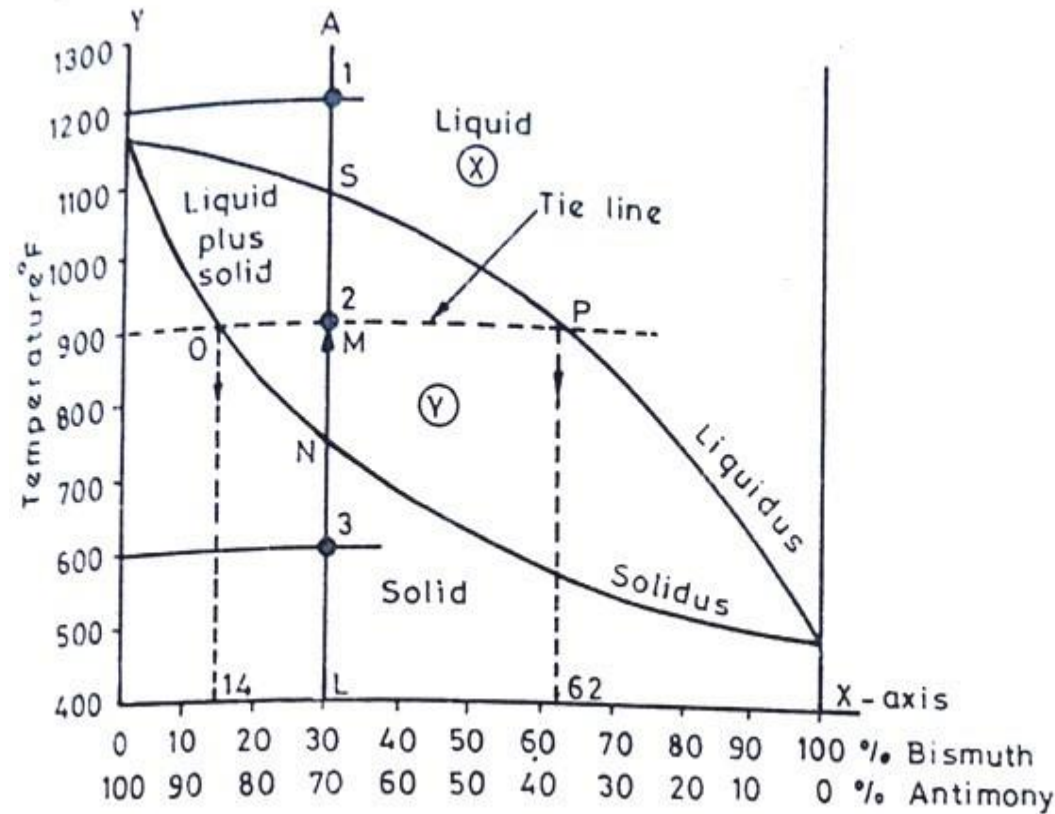


Rule -2 : Phase Composition

- To find out the composition of phases which are stable at given temp. (say 900°F), draw a horizontal line, OP at the given temp.
- The projections of the intersections of the isothermal line with the solidus and liquidus respectively, give the compositions of the solid and liquid, which co-exist in equilibrium at the temp.
- Liquid phase (point – P) has the composition roughly 62% bismuth.
- Solid phase (point - O) has the composition roughly 14% bismuth.



Rule -3 : Lever Arm Principle



- Beside indicating the number of phases and phase composition the phase diagram also tells the proportion of co-existing phases at any given temp.
- To determine the relative amount of two phases, erect an ordinate at a point (say 30% Bi) on the composition scale which give the total or overall composition of the alloy.

Rule -3 : Lever Arm Principle

- The intersection of this composition vertical (AL) and a given isothermal line OP (i.e., Point M) is the fulcrum of a simple lever system and OM and MP are two lever arms, the relative lengths of the lever arms multiplied by the amounts of the phase present must balance.
- The amount of a given phase multiplied by its lever arm is equal to the amount of the other phase multiplied by its (i.e., Other) lever arm this is called the lever rule.

Rule -3 : Lever Arm Principle

- It can also be seen that the proportion of solid corresponds to the length of the segment adjacent to liquidus line, whereas the fraction of liquid corresponds to the length of segment adjacent to the solidus line.
- The isotherm (line OMP) can be considered as a tie line, since it joins the composition of two phases in equilibrium at a specific temperature.
- The lever rule or principle may be expressed mathematically as:

1. The amount of solid phase

$$= \frac{MP}{OP} \times 100 = \frac{(62 - 30)}{(62 - 14)} \times 100 = 66\frac{2}{3}\%$$

2. The amount of liquid phase

$$= \frac{OM}{OP} \times 100 = \frac{30 - 14}{62 - 14} \times 100 = 33\frac{1}{3}\%.$$

Gibb's Phase Rule

- Gibbs phase rule establishes the relationship between the number of variable (F), the number of element (C), and the number of phases(P).
- It is expressed mathematically as follows:

$$P + F = C + 2$$

Where, P = Number of phases in system

F = Number of variables that can be changed independently without effecting number of phases

C = Number of elements

'2' represent any two variables amongst temperature, pressure and composition

Gibb's Phase Rule

- In general all equilibrium diagram studied at constant pressure, hence Gibb's phase rule is modified to

$$P + F = C + 1$$

- Phase rule helps to determine maximum number of phase present in an alloy system under equilibrium conditions at any point in phase diagram.
- The phase rule can also be used to determine the degree of freedom that can be changed

Hume - Rothery's Rules For Solid Solubility

- Solid solution is an alloy of two or more element wherein the atomic crystal structure of alloying element (solute) is same as that of the base metal matrix (solvent).
- The solubility limit of the solute in the solvent(of the alloying element in base metal matrix) is governed by certain factors.
- These governing factors are known as Hume -Rothery's rules for solid solubility.

Hume - Rothery's Rules For Solid Solubility

These governing factor are as follows.

1. Atomic size:

- Alloying elements having similar atomic size as that of the base metal matrix have better solid solubility.
- For a favorable solid solution formation, the difference of atomic size of solute and solvent should be less than 15 %.

2. Chemical affinity:

- Element having lower chemical affinity have greater solid solubility.
- Element having higher chemical affinity have the tendency of formation of compound and hence restrict formation of solid solution.
- In general, the alloy elements located closer in the periodic table have higher solid solubility.

Hume - Rothery's Rules for Solid Solubility

3. Relative valency:

- Metals having lower valency have more solubility for metals having higher valency.
- Hence, for better solubility, the base metal selected should be one that has lower valency as compared to that of alloying elements.

4. Crystal structure:

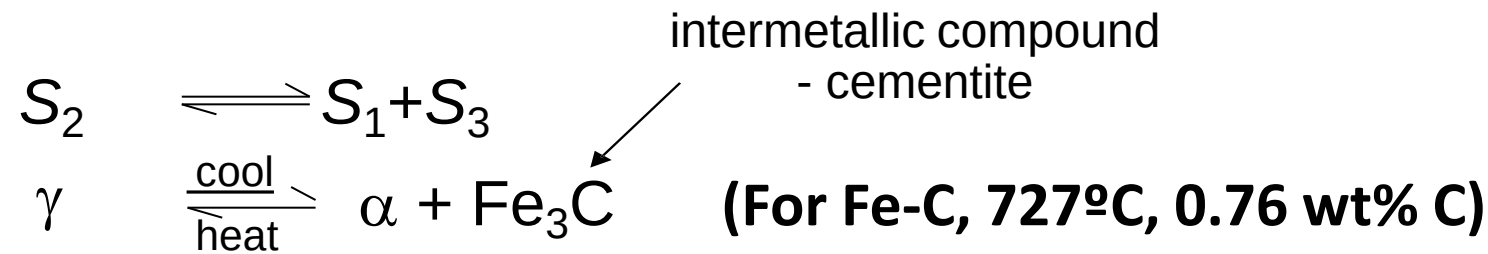
- As mentioned earlier, solid solution is an alloy of element having similar crystal structure.
- Difference in crystal structure limits the solid solubility of elements.

Eutectic, Eutectoid And Peritectic Reactions

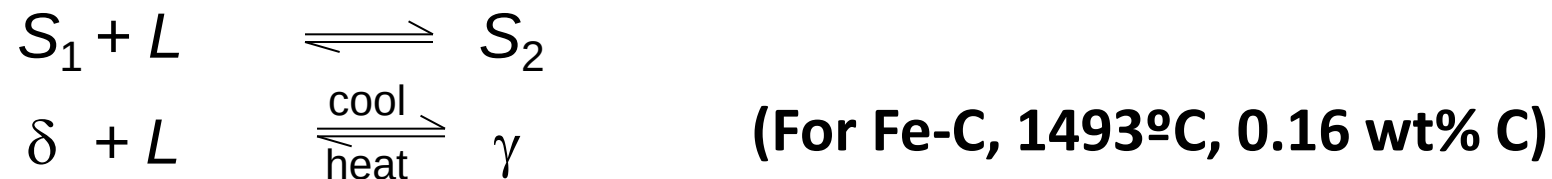
- Eutectic - liquid transforms to two solid phases



- Eutectoid – one solid phase transforms to two other solid phases



- Peritectic - liquid and one solid phase transform to a second solid phase

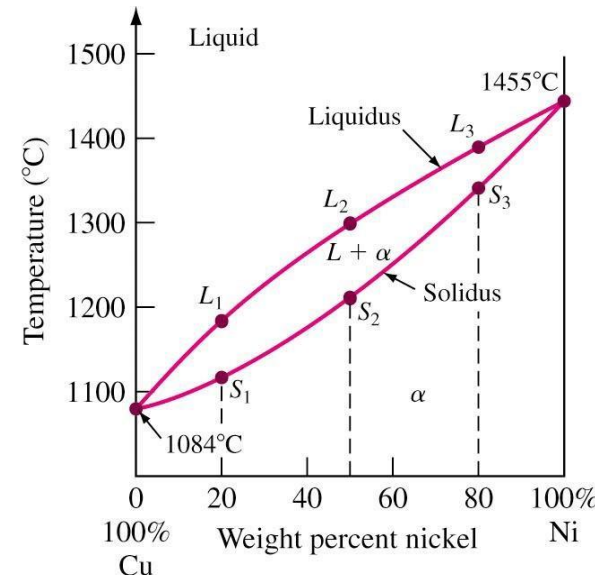
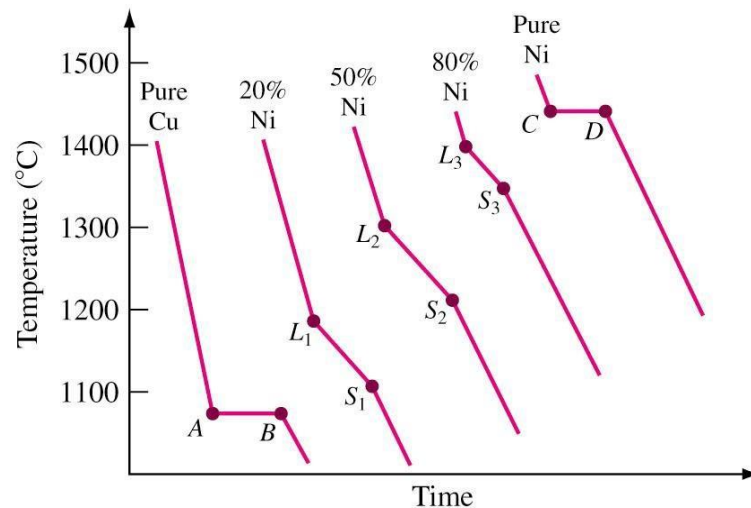


Series of cooling curves with points corresponding to solid-liquid transition for different metal composition are first constructed.

Points of change of slope of cooling curves (thermal arrests) are noted and phase diagram is constructed by joining these solid-liquid transition temperatures.

More the number of cooling curves, more accurate is the phase diagram.

- Curve joining solidification start-liquidus line
- Curve joining solidification ends—solidus line



- Phase changes can be analyzed:-
- 1) completely soluble in the liquid state and the solid state.
- 2) For two metals completely soluble in liquid state and insoluble in the solid state
- 3) For two metals completely soluble in liquid state and partially soluble in the solid state

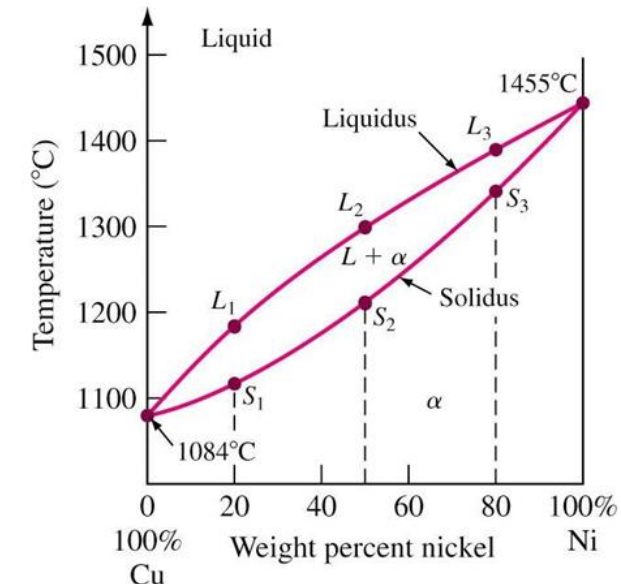
● In alloys, when two components are mixed, they could be completely soluble in the liquid state and the solid state. Such a binary system

● ~~Cu-Ni, Ag-Cu, Ge-Si~~ etc., are examples

● Cu-Ni is the simplest among to understand.

● Composition of Ni ranges from 0%100% from left to right and *vice versa* for copper composition.

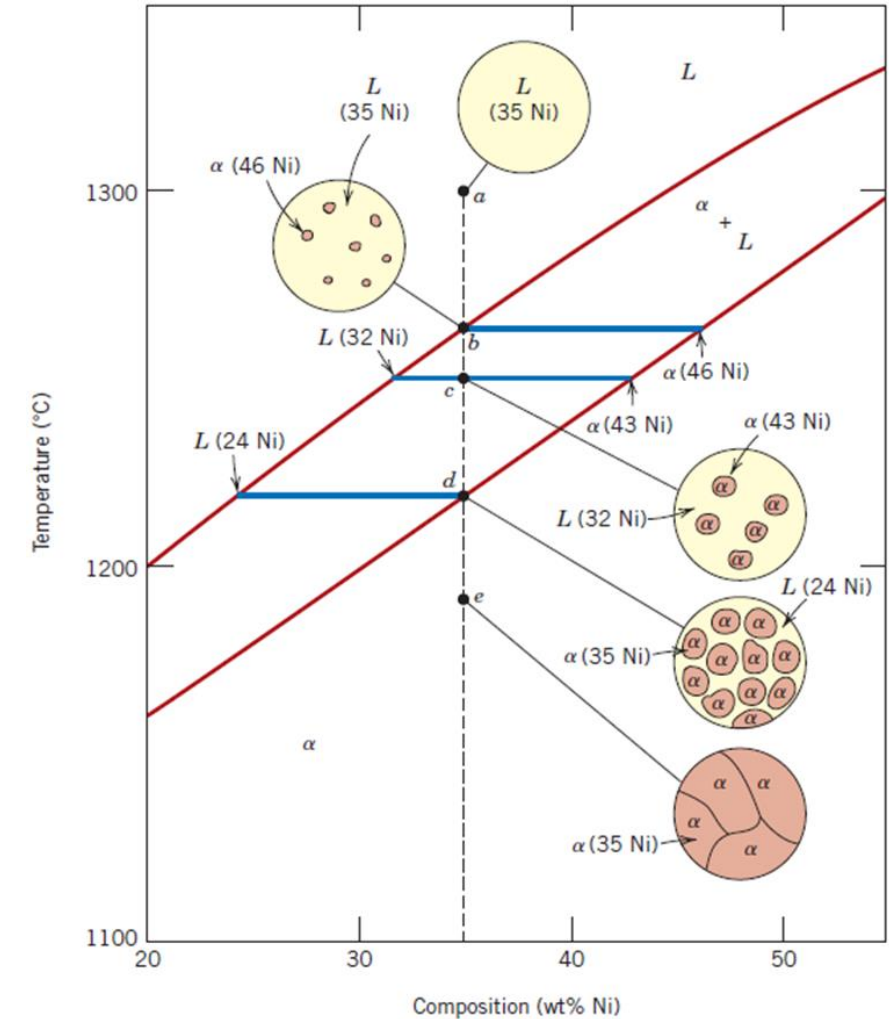
● There are three distinct phases which are liquid, α (solid) phase and ($\alpha+L$)



Development of microstructure in isomorphous alloys

Equilibrium Cooling

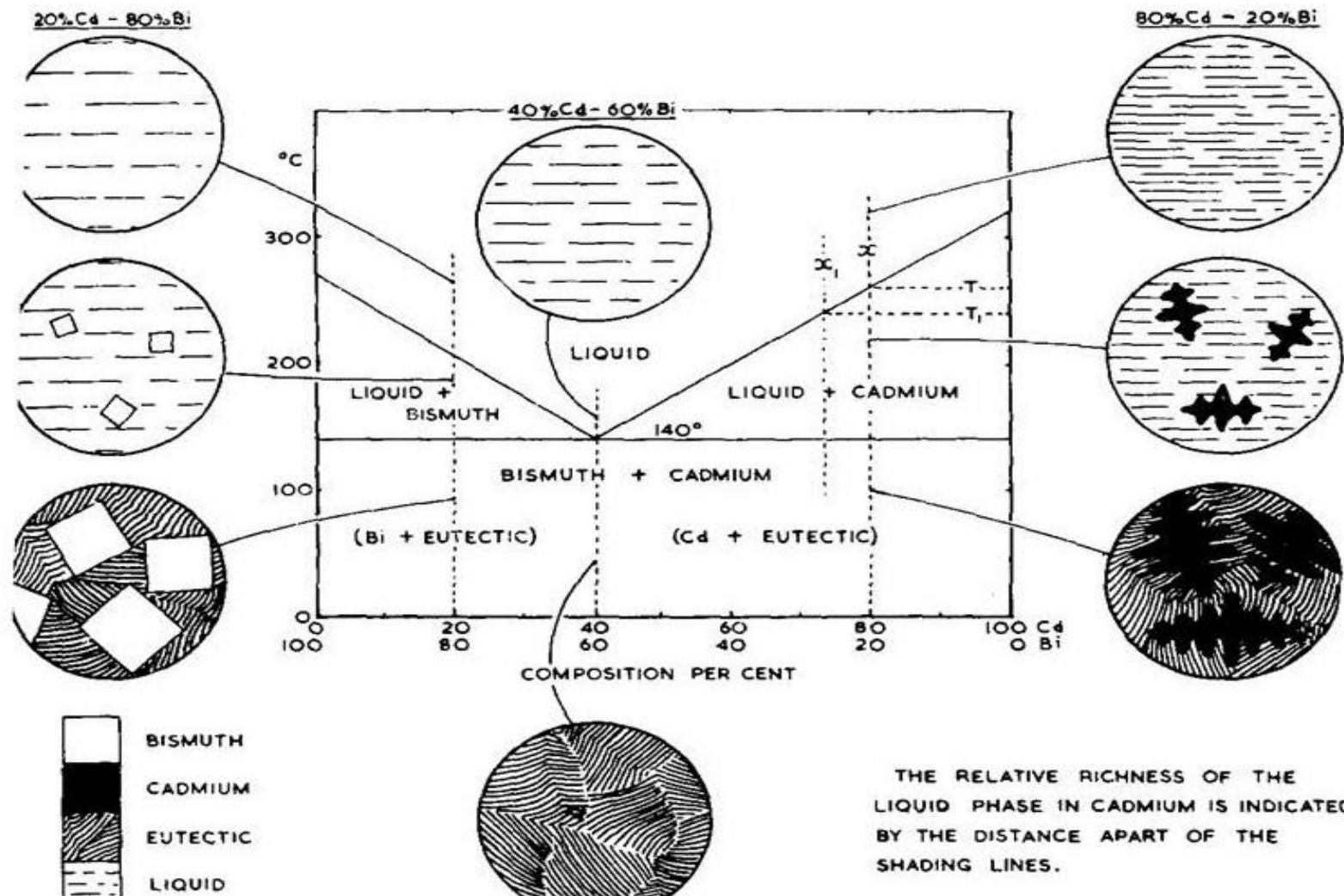
- During equilibrium, the cooling occurs very slowly, and phase equilibrium is continuously maintained.
- Consider the copper–nickel system of composition 35 wt% Ni–65 wt% Cu; and it is cooled from 1300°C.



- Cooling of an alloy of the above composition corresponds to moving down the vertical dashed line.
- At point *a*, the alloy is completely liquid (of composition 35 wt% Ni-65 wt% Cu) and has the microstructure represented by the circle inset in the figure.
- As cooling begins, no microstructural or compositional changes will be present until we reach the liquidus line (point *b*,).
- At point *b*, the first solid begins to form, which has a composition dictated by the tie line drawn at this temperature [i.e., 46 wt% Ni-54 wt% Cu, noted as (46 Ni)];
- The composition of liquid is still approximately 35 wt% Ni-65 wt% Cu [*L*(35 Ni)], which is different from that of the solid .
- With continued cooling, both compositions and relative amounts of each of the phases will change. The compositions of the liquid and phases will follow the liquidus and solidus lines, respectively.
- Furthermore, the fraction of the phase will increase with continued cooling. Note that the overall alloy composition (35 wt% Ni-65 wt% Cu) remains unchanged during cooling even though there is a redistribution of copper and nickel between the phases.
- At point *c* in Figure 9.4, the compositions of the liquid and phases are 32 wt% Ni-68 wt % Cu [*L*(32 Ni)] and 43 wt% Ni-57 wt% Cu [(43 Ni)], respectively.
- The solidification process is virtually complete at about point *d*; the composition of the solid is approximately 35 wt% Ni-65 wt% Cu (the overall alloy composition) while that of the last remaining liquid is 24 wt% Ni-76 wt% Cu.
- Upon crossing the solidus line, this remaining liquid solidifies; the final product then is a polycrystalline -phase solid solution that has a uniform 35 wt% Ni-65 wt% Cu composition (point *e*, Figure 9.4).
- Subsequent cooling will produce no microstructural or compositional alterations.

Phase Changes In Alloys

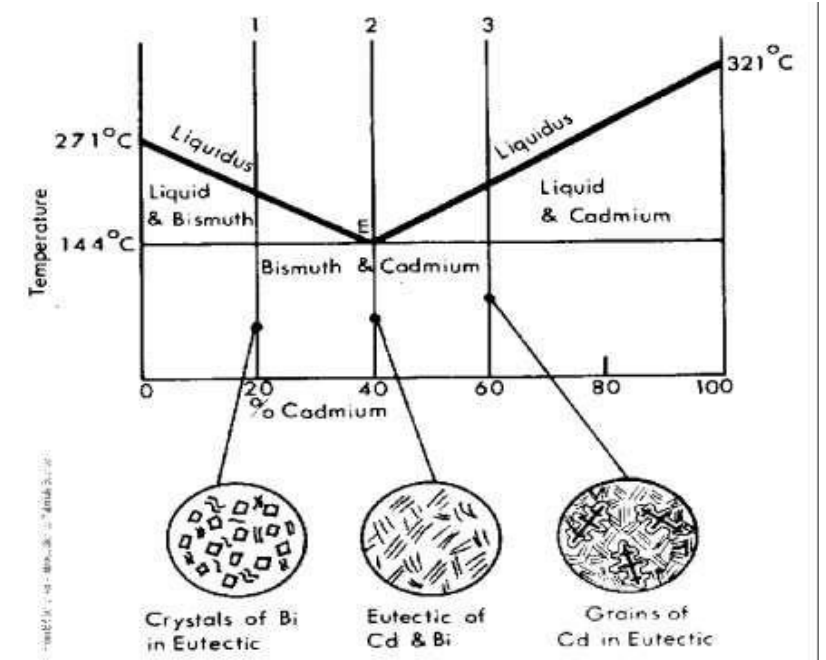
- ❖ For two metals completely soluble in liquid state and insoluble in the solid state
- Consider the Bismuth Cadmium system as shown :



Phase Changes In Alloys

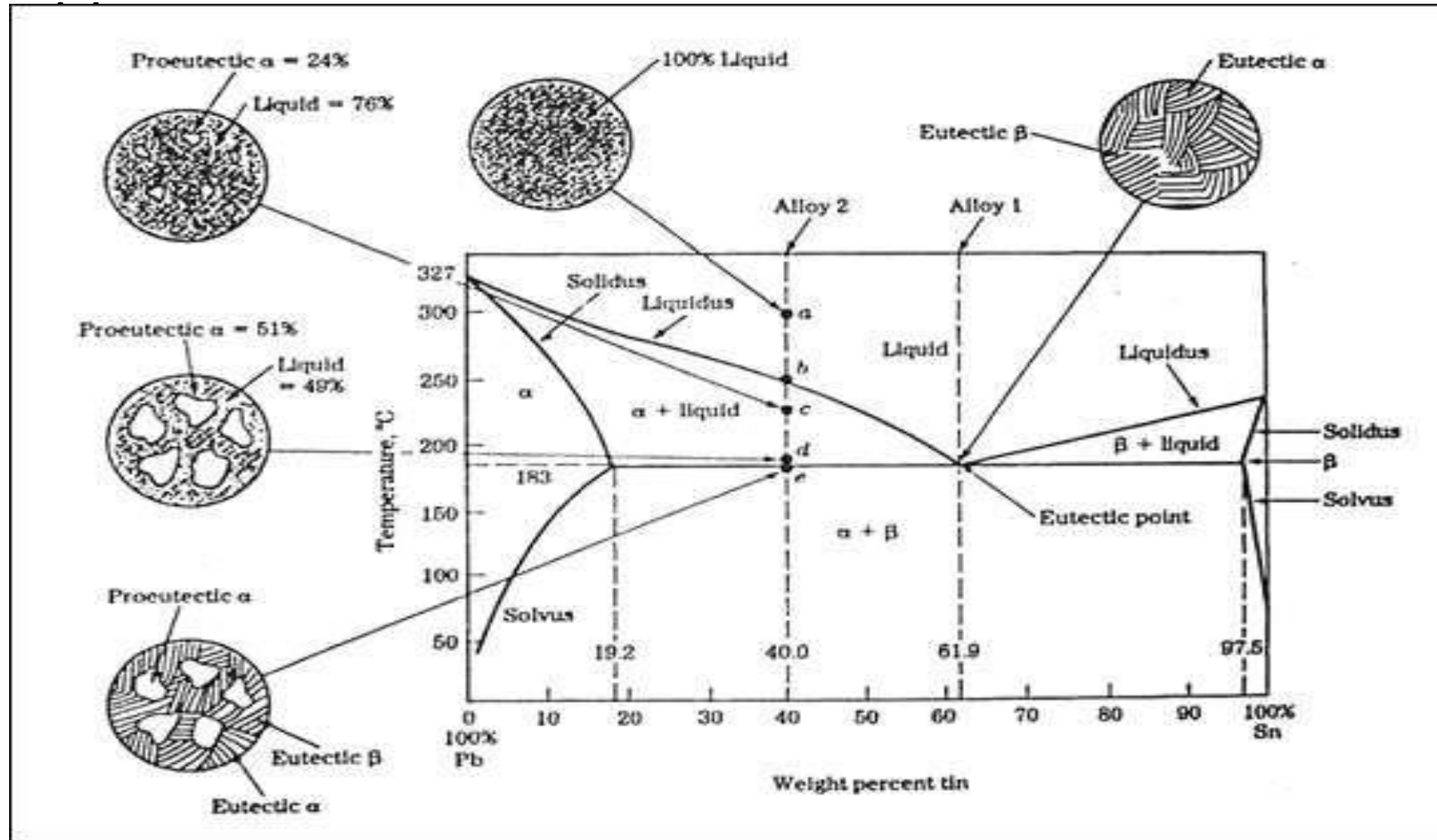
Four phase fields are seen on figure :

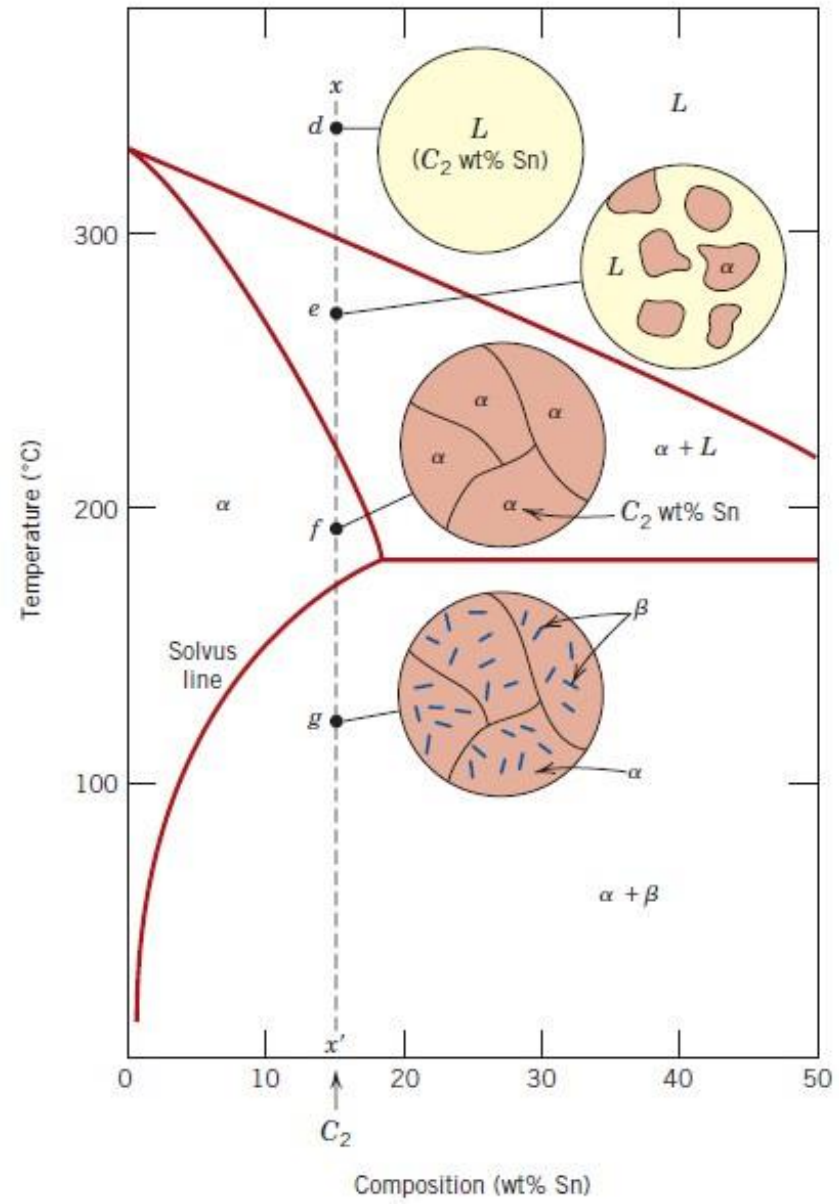
- ✓ a fully molten region,
- ✓ two regions in which liquid and solid coexist,
- ✓ a completely solid region.
- The two liquidus lines slope downwards from the melting points of the pure materials to meet at a point known as the eutectic point (E).
- Material of the eutectic composition will undergo the transformation liquid $\rightarrow$ solid (Bi + Cd) at the temperature 144° C.
- Since the formation of the two solids occurs simultaneously, this will be also reflected in the microstructure.
- The two solids are strongly mixed in this eutectic structure, sometimes as particles or "stars" of one solid surrounded by the other, or maybe as laminations as shown in the drawing.
- Note that unlike all other alloys, eutectic compositions have no extended freezing range but they freeze at a definite temperature similarly to pure metals.

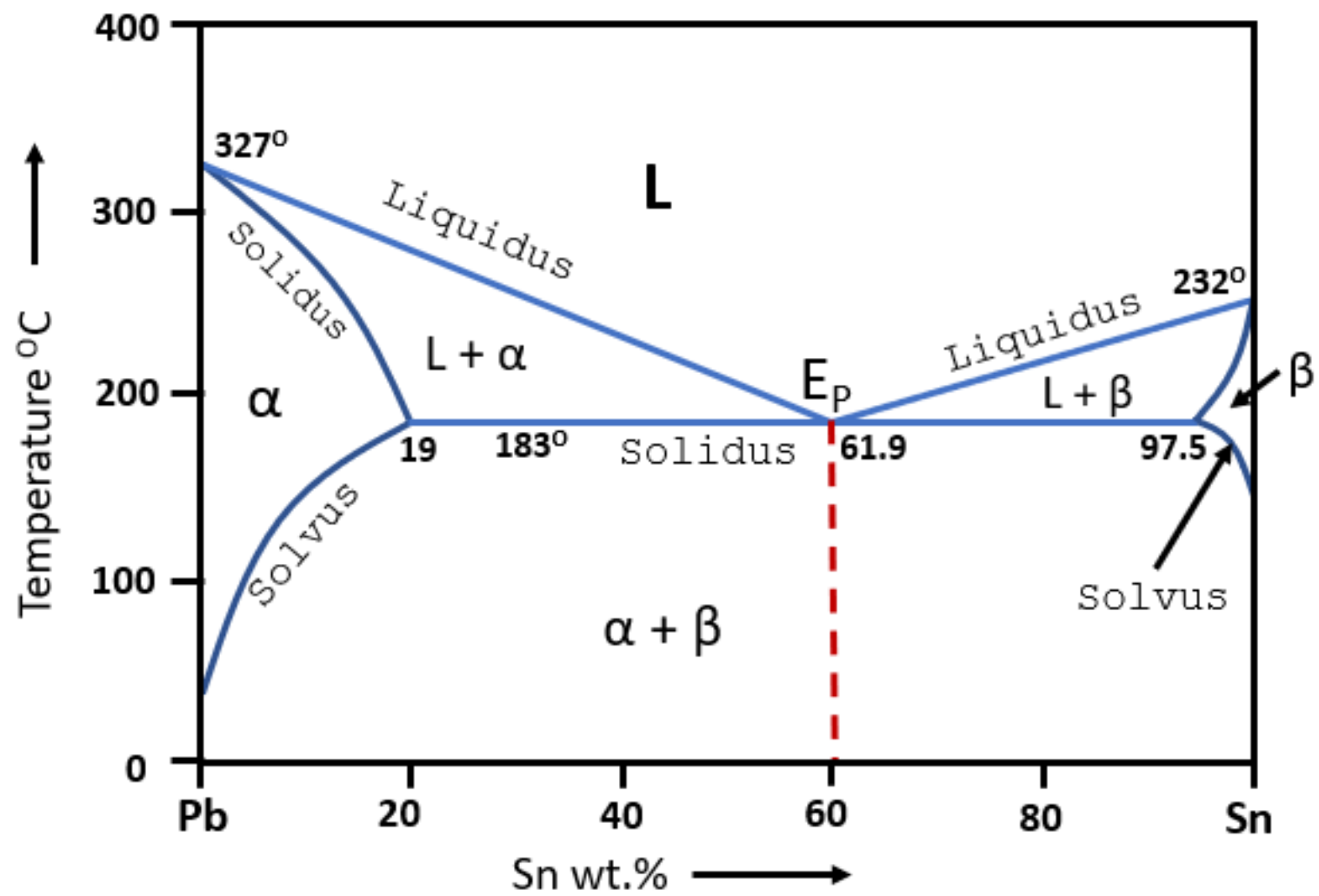


Phase Changes In Alloys

- ❖ For two metals completely soluble in liquid state and partially soluble in the solid







Phase Changes In Alloys

- A eutectic system can occur when terminal solid solutions exist on both ends of the binary equilibrium phase diagram. An example of a binary eutectic system is lead (Pb) - tin (Sn).
- Although the atomic size difference is less than 10%, Pb has an FCC crystal structure while Sn is an unusual metal with a non-cubic tetragonal structure.
- This results in limited solid state solubility with the maximum solubility of Sn in the FCC Pb equal to 19.2 wt%Sn while only 2.5wt% of Pb is soluble in the tetragonal Sn structure.
- At compositions and temperatures, which exceed these solubility limits, two solid phases will exist in equilibrium.
- The phase is the FCC Pb with some substitutional Sn atoms and the phase is tetragonal Sn with only a few substitutional Pb atoms.
- These maximum solid-state solubility both occur at 183 deg. Cel. which is referred to as the eutectic temperature.
- At this temperature, there exists a point on the phase diagram (a single combination of composition and temperature) where three phases (the two solids and a liquid) can exist simultaneously in equilibrium.

PLATINUM SILVER PHASE DIAGRAM

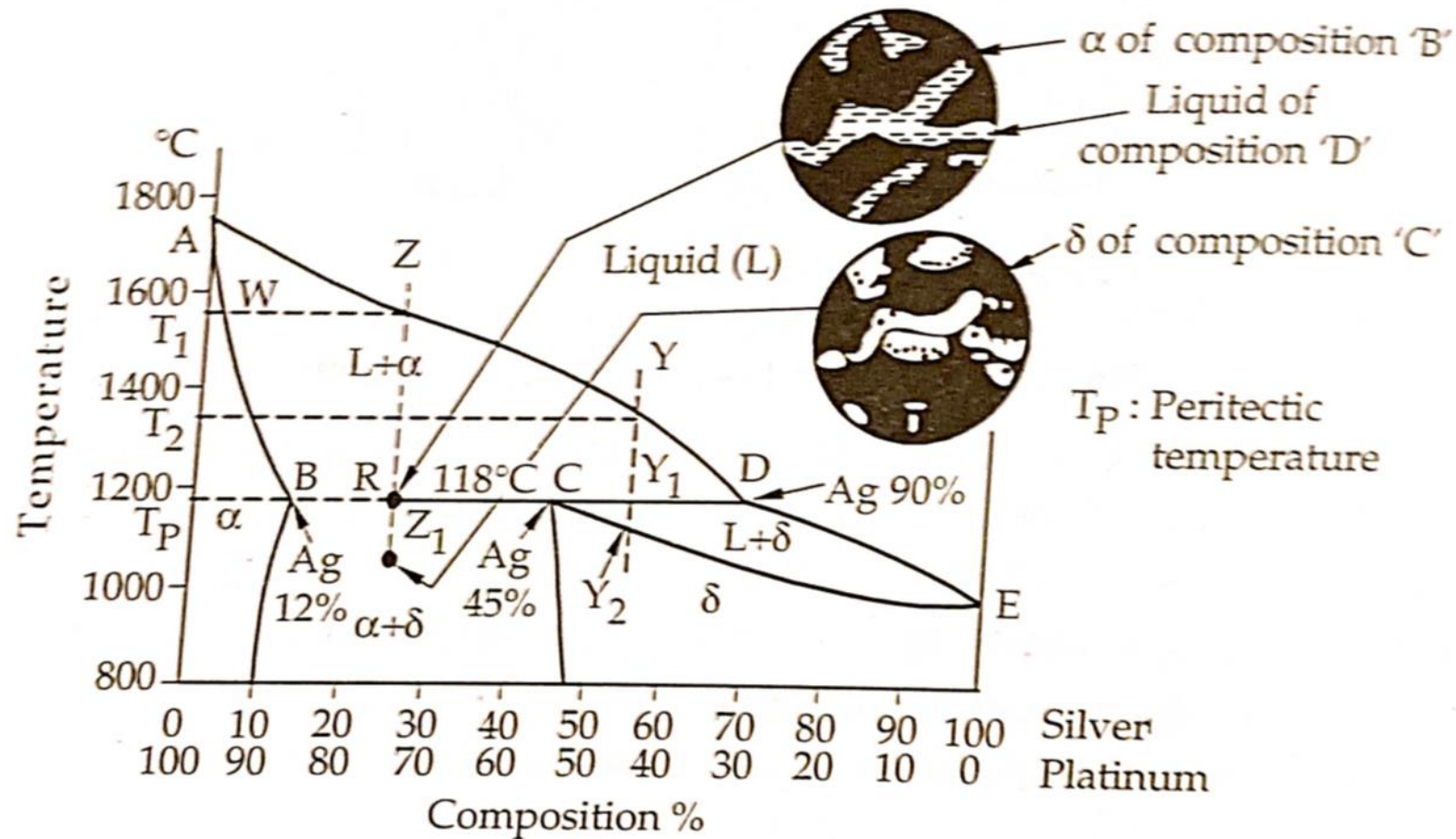


Fig. 39.8 Peritectic Reaction (Platinum-Silver Equilibrium Diagram)

Cooling Curves

- A mathematical curve while plotting the temperature as a function of time as different alloys in the system are very slowly cooled.
- This method is basically used to determine the temperature at which phase change (from liquid to solid) occurs.
- The study is useful in :-
 - Studying the changes that occur during solidification of alloys, and
 - In determining transformations subsequent to solidification .